Modeling and Analysis of Airflow in the Volvo Bluff-Body Experiment using OpenFOAM: Turbulence Model Validation and Proper Orthogonal Decomposition

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Problem Statement:

To numerically simulate airflow around a bluff-body (specifically, a prism) based on the Volvo bluff-body burner experiment. The focus is on the non-reacting flow case with an inlet velocity of 16.6 m/s. The primary objectives are:

1. To evaluate which turbulence model implemented in OpenFOAM best replicates the experimental data.
2. To explore Adaptive Mesh Refinement (AMR) for improved spatial resolution in flow regions of interest.
3. To apply Proper Orthogonal Decomposition (POD) to extract dominant flow structures based on their energy content.

Volvo Bluff-Body Burner Overview

- Features a canonical bluff-body stabilized premixed flame configuration.
- Used to study flame anchoring and instabilities under lean combustion conditions.

Lean combustion:

- Desirable for high efficiency and low emissions.
- Prone to flame instability and lean blowout (LBO) risks.
- Studied experimentally under the Volvo Flygmotor AB program.

Experimental Data:

- Laser Doppler Anemometry (LDA)
- Coherent Anti-Stokes Raman Spectroscopy (CARS)

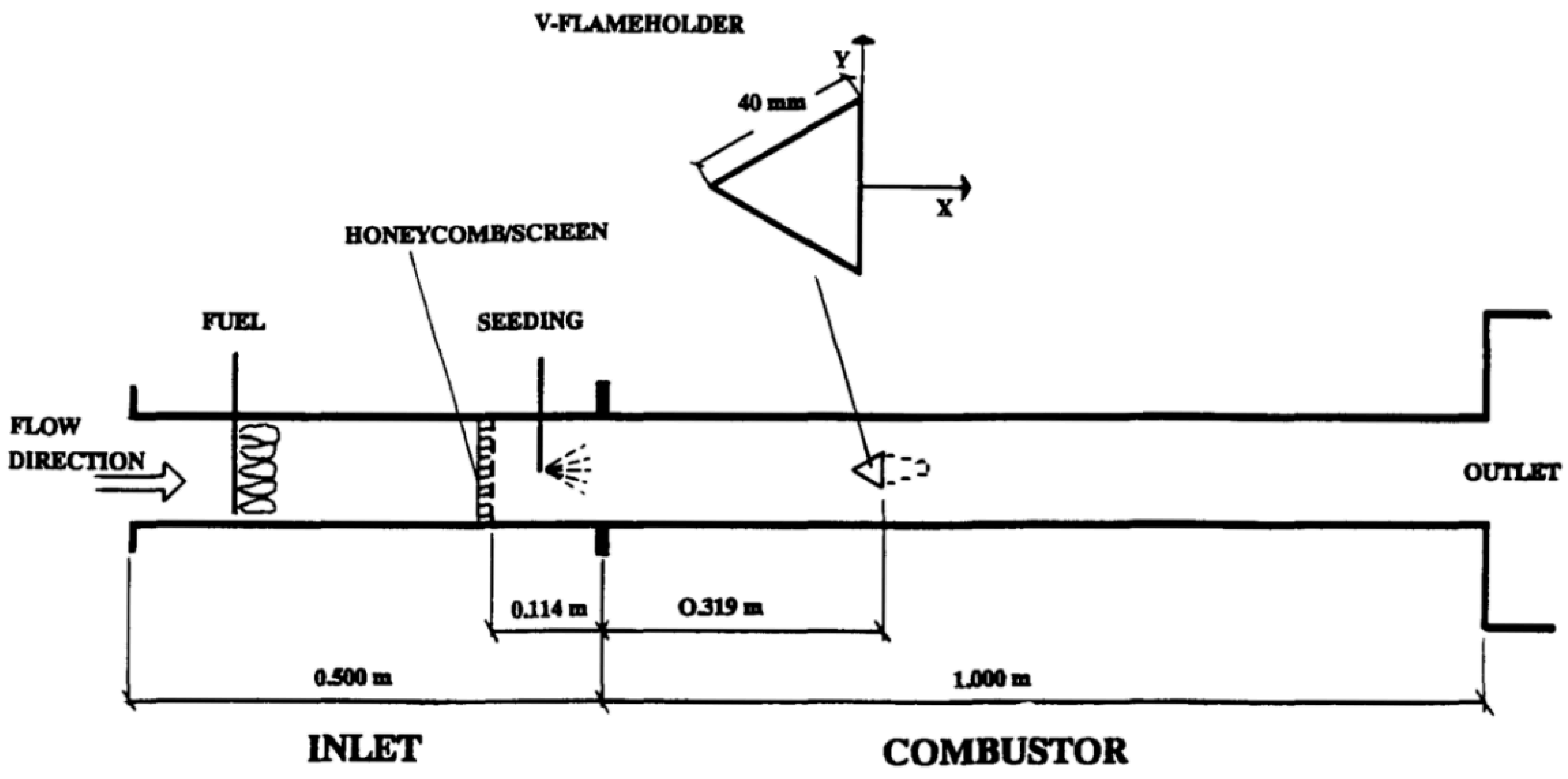


Figure 1: Schematic of the Volvo test rig (Sjunneson et al. [3])

Problem Statement:

Re= 48,000
Pout(gauge) =0
U_{bulk}[m/s]= 16.6

Equivalence Ratio (ϕ)= 0
T_{in} [K]= 288

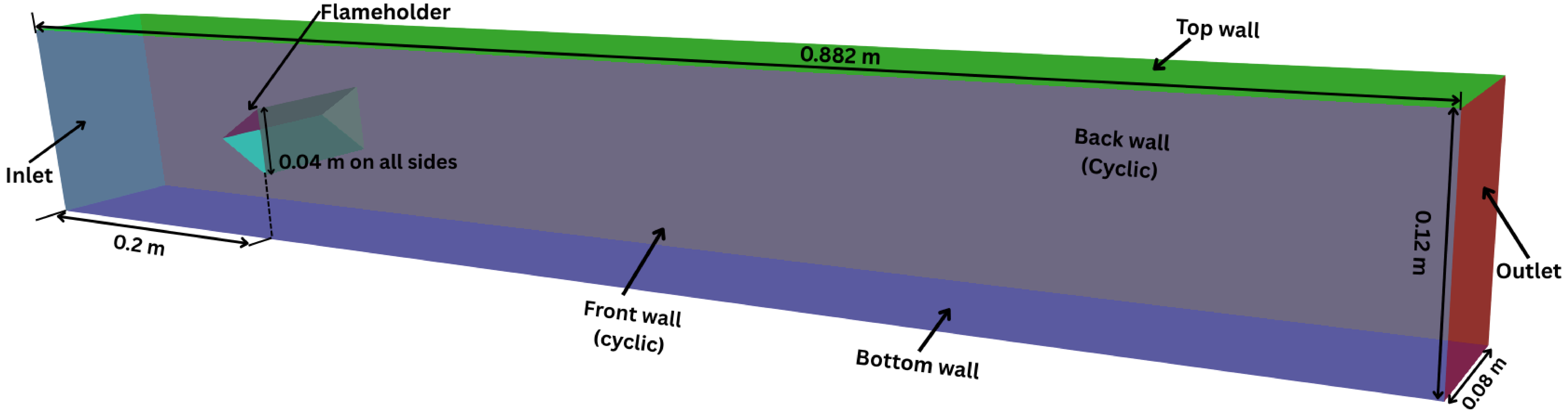


Figure 2 : Computational domain and boundary Conditions

Meshing

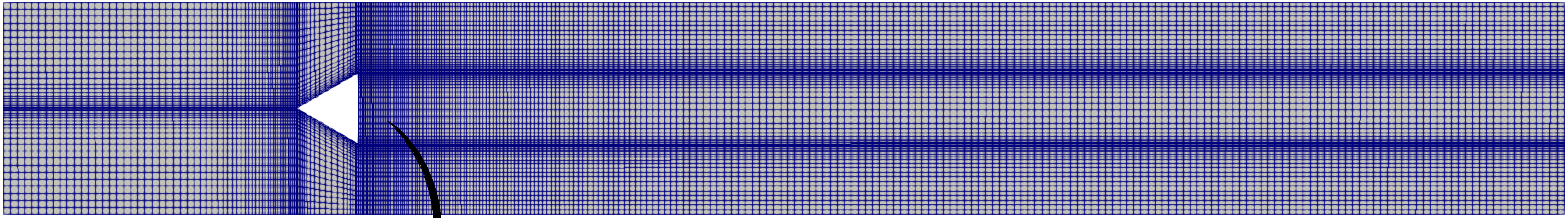
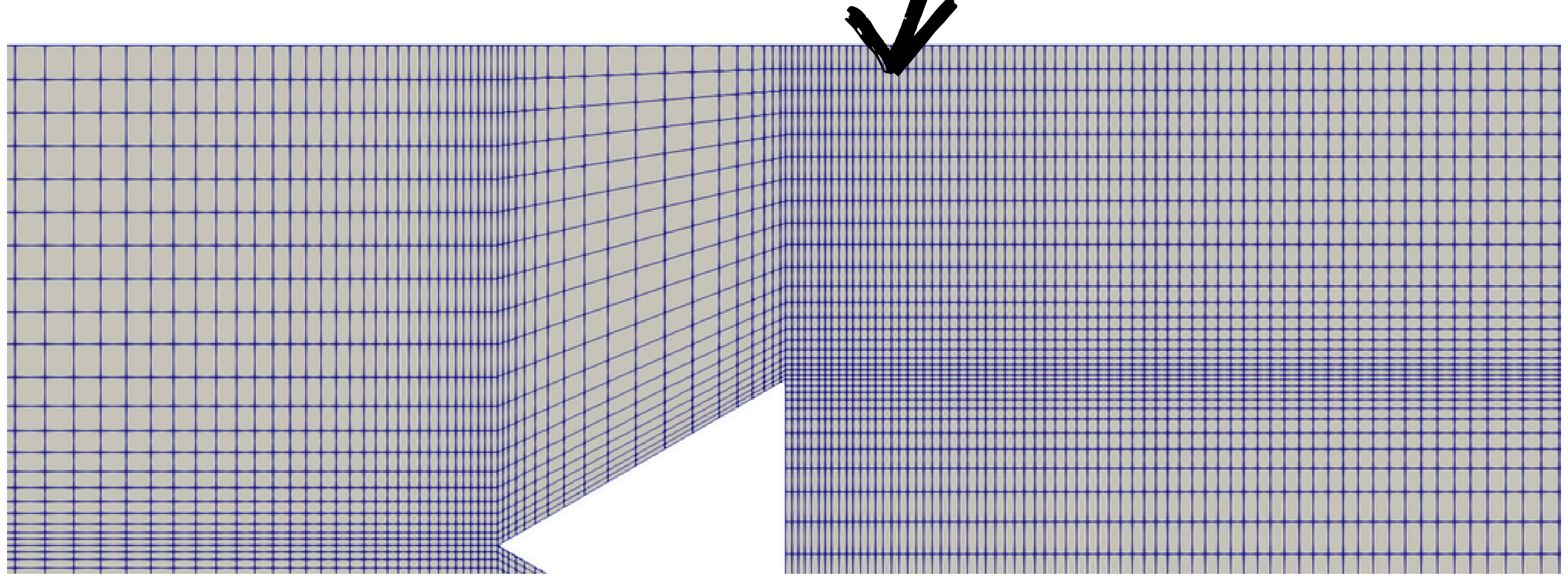


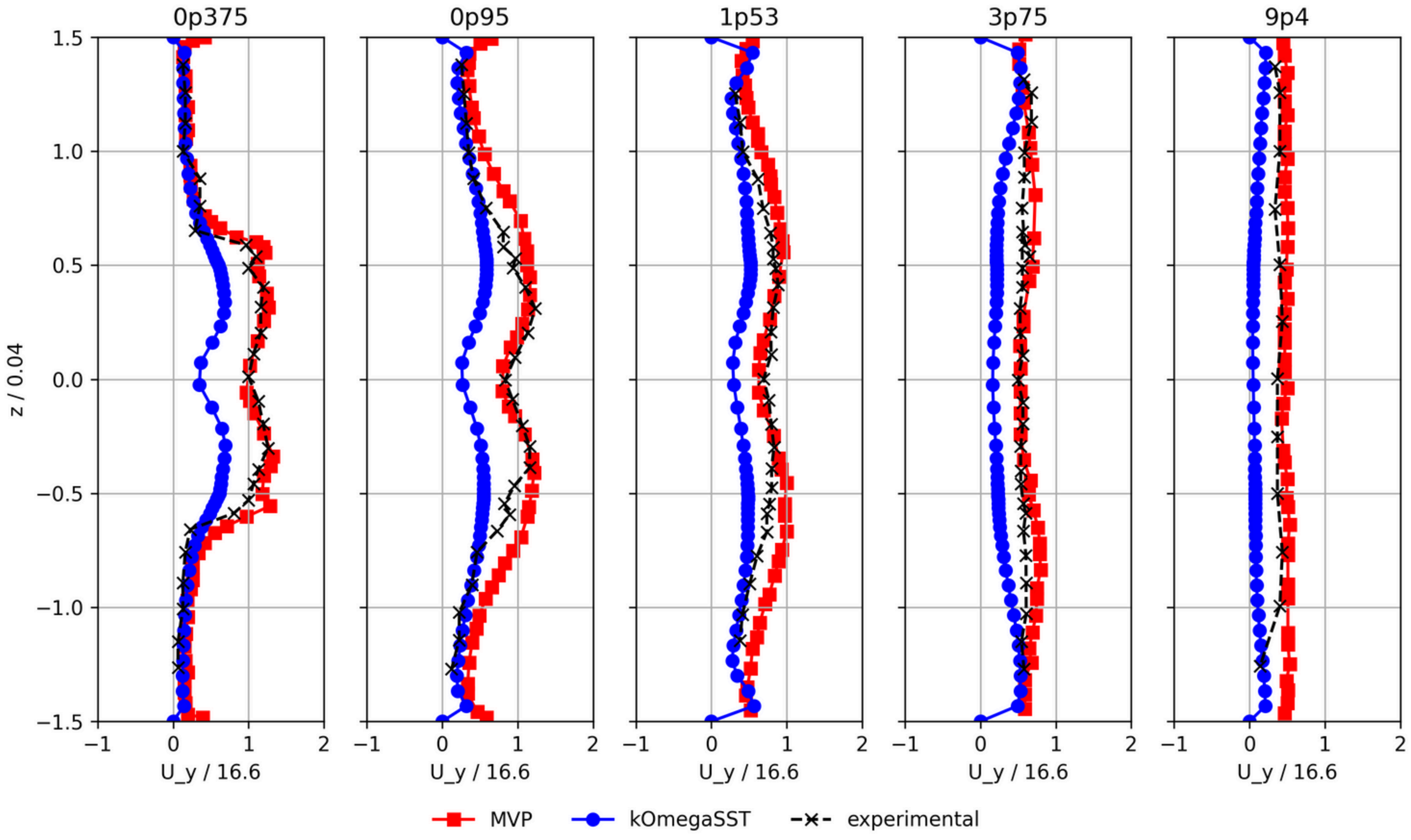
Figure : Structured Mesh

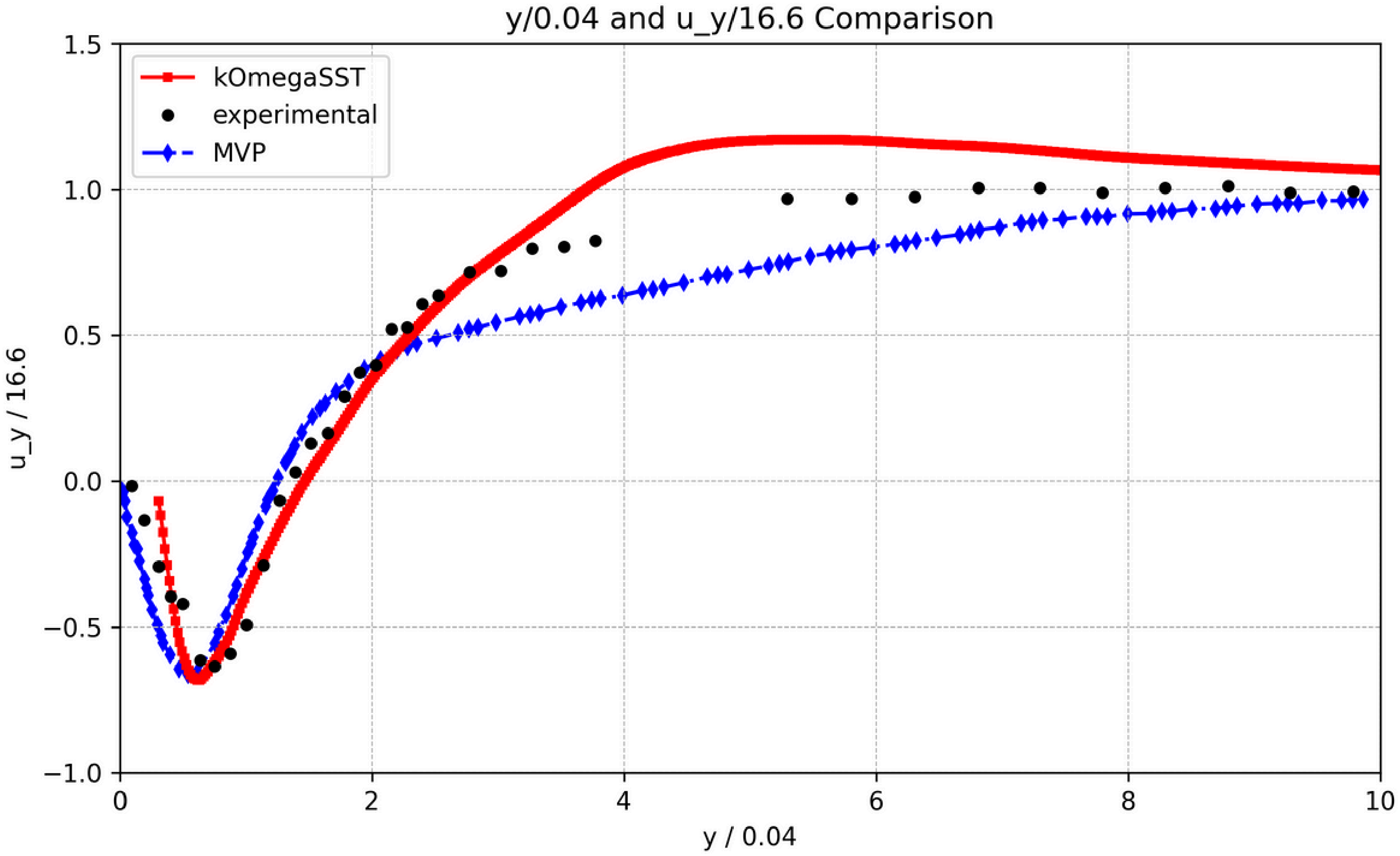


No. of Cells : 535,928 (Hexahedral Only)

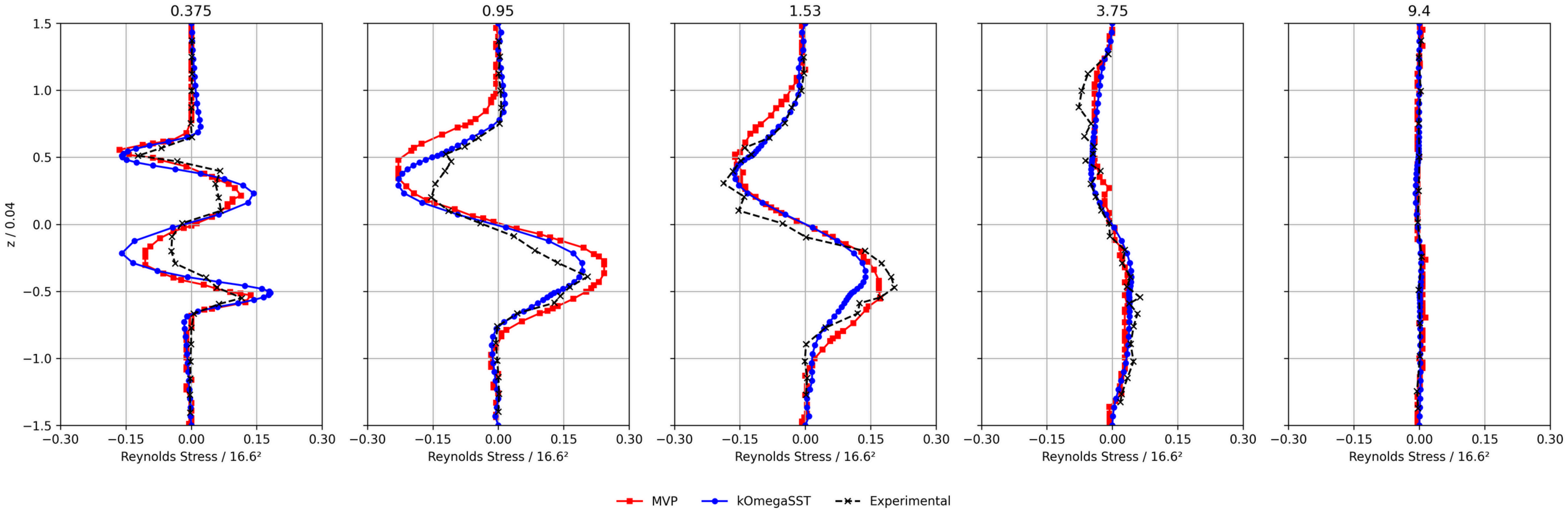
No. of nodes : 565,536

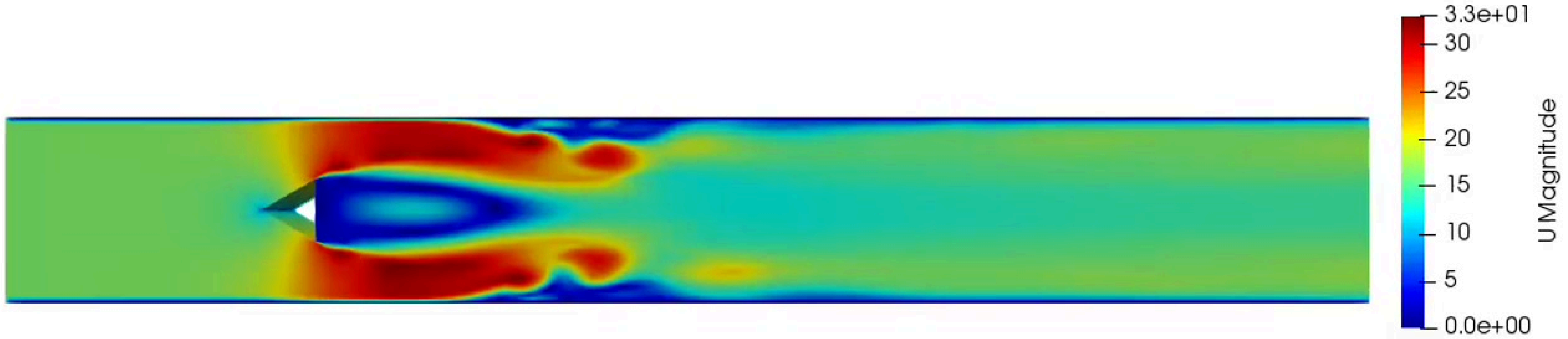
Results(K Omega SST)



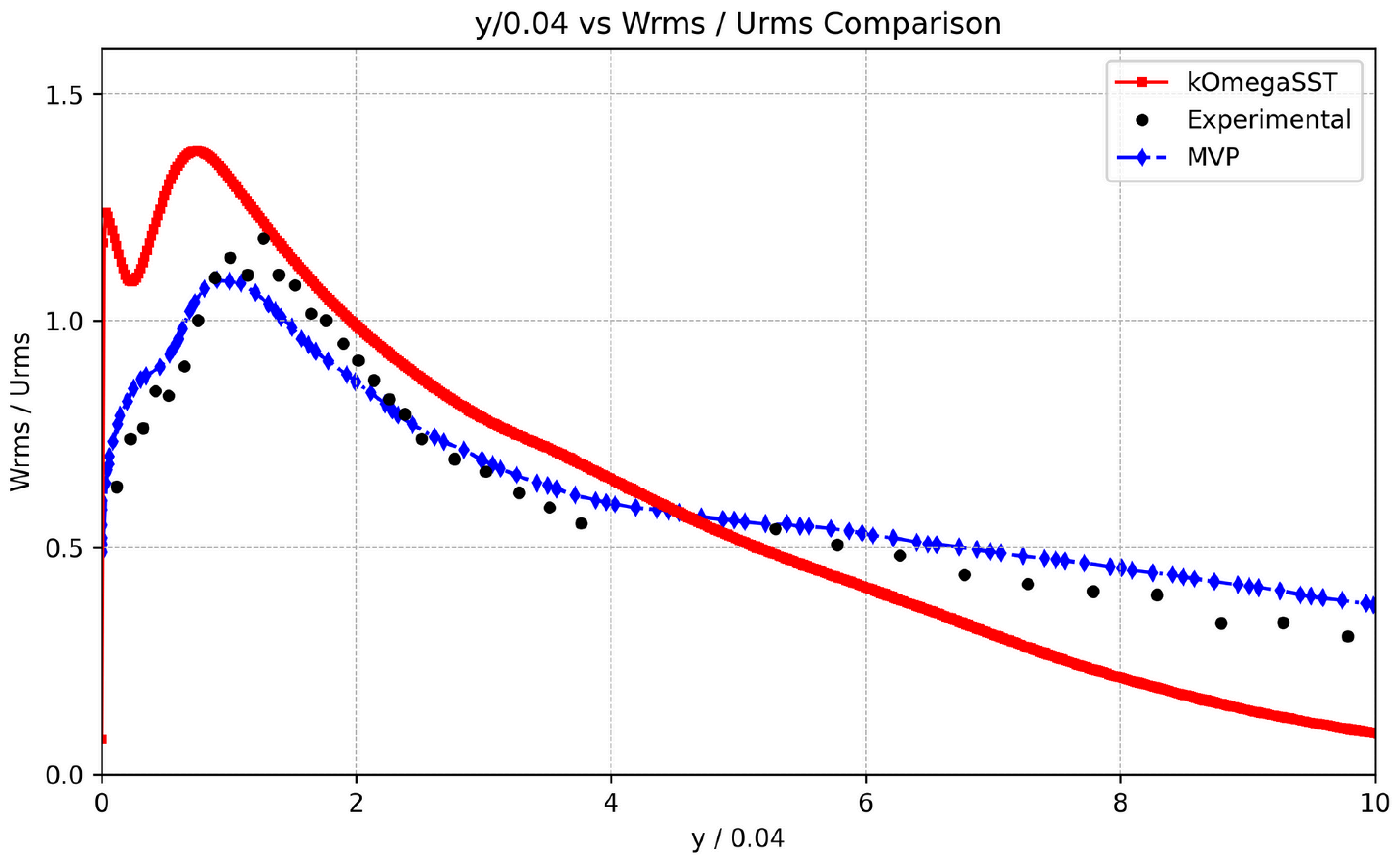


Results(K Omega SST)

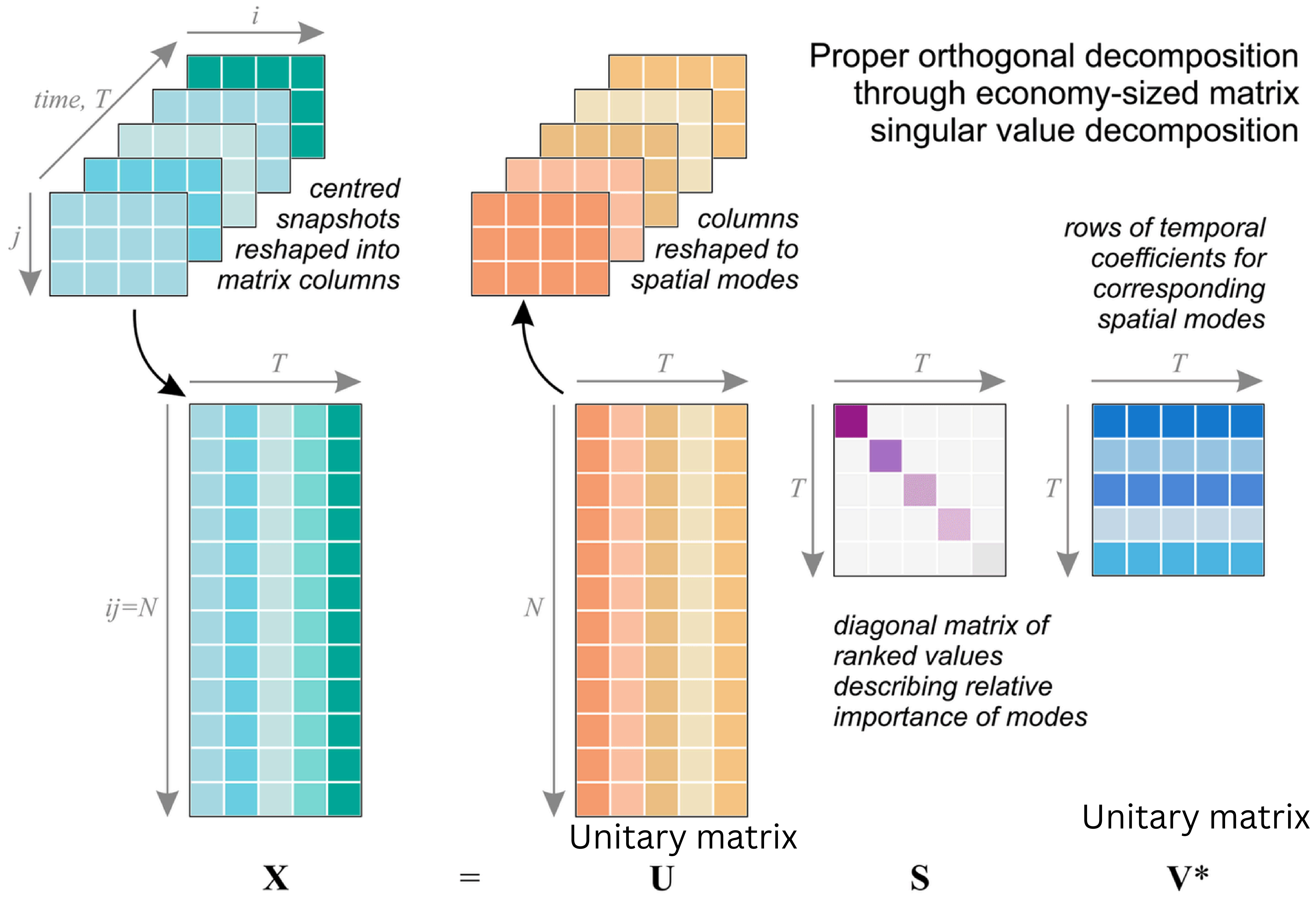




Results(K Omega SST)



Proper Orthogonal Decomposition (POD) :



Proper Orthogonal Decomposition (POD) :

POD identify the most energetic structures within the simulation and quantify their contribution to the overall flow behavior.

Goal:

- Analyze flow field data from OpenFOAM simulations.
- Extract dominant flow structures using POD.
- Save modes and amplitudes for reconstruction or reduced-order modeling.

Procedure:

- convert the termporal and spatial data into matrix form
- flatterned the spatial data into columns
- each column represent a snapshot of time
- Decompose the matrix with high order into three matrix - S, U, V*

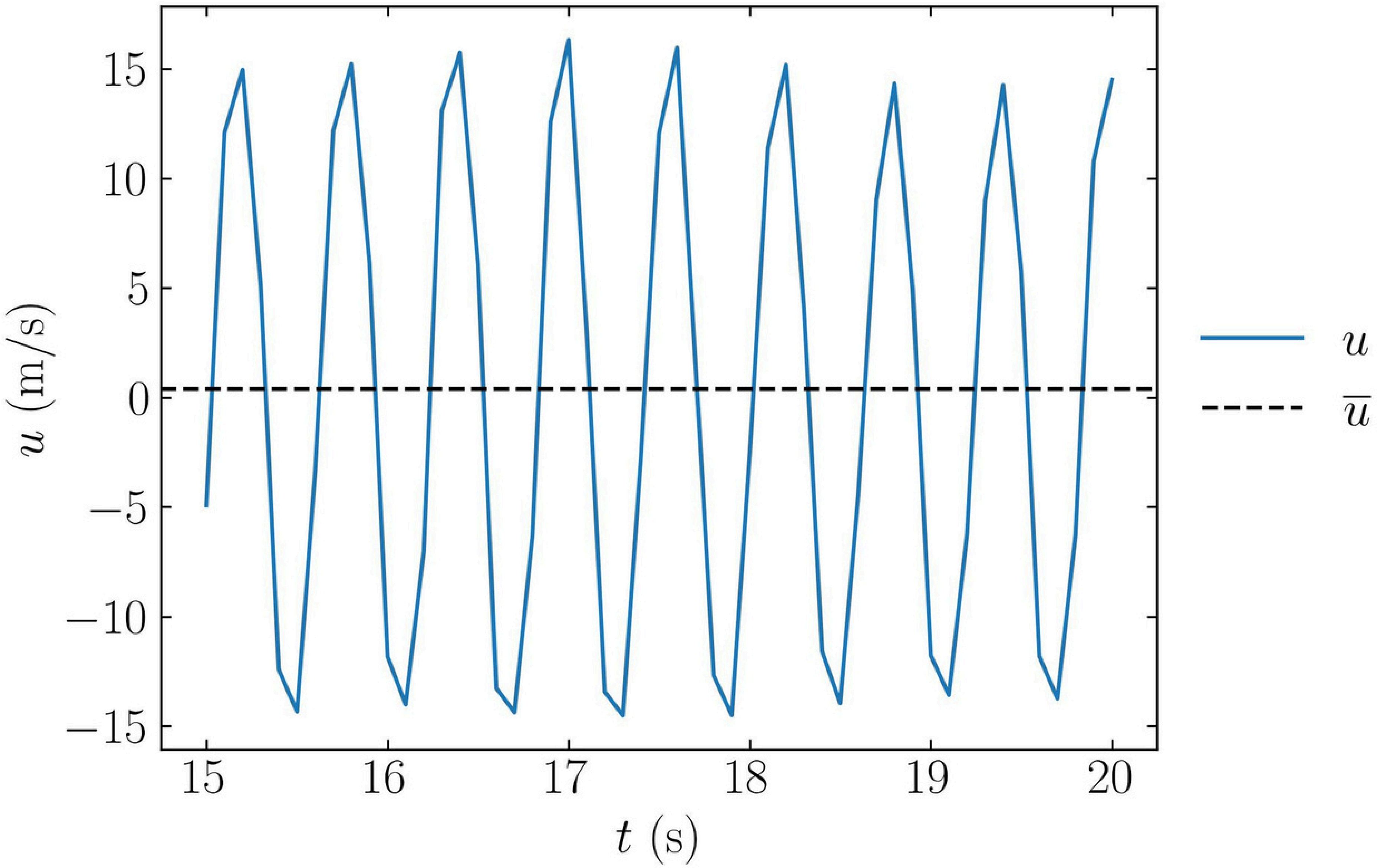
Fast Fourier Transform(FFT):

FFT is Used to find vortex shedding cycle time

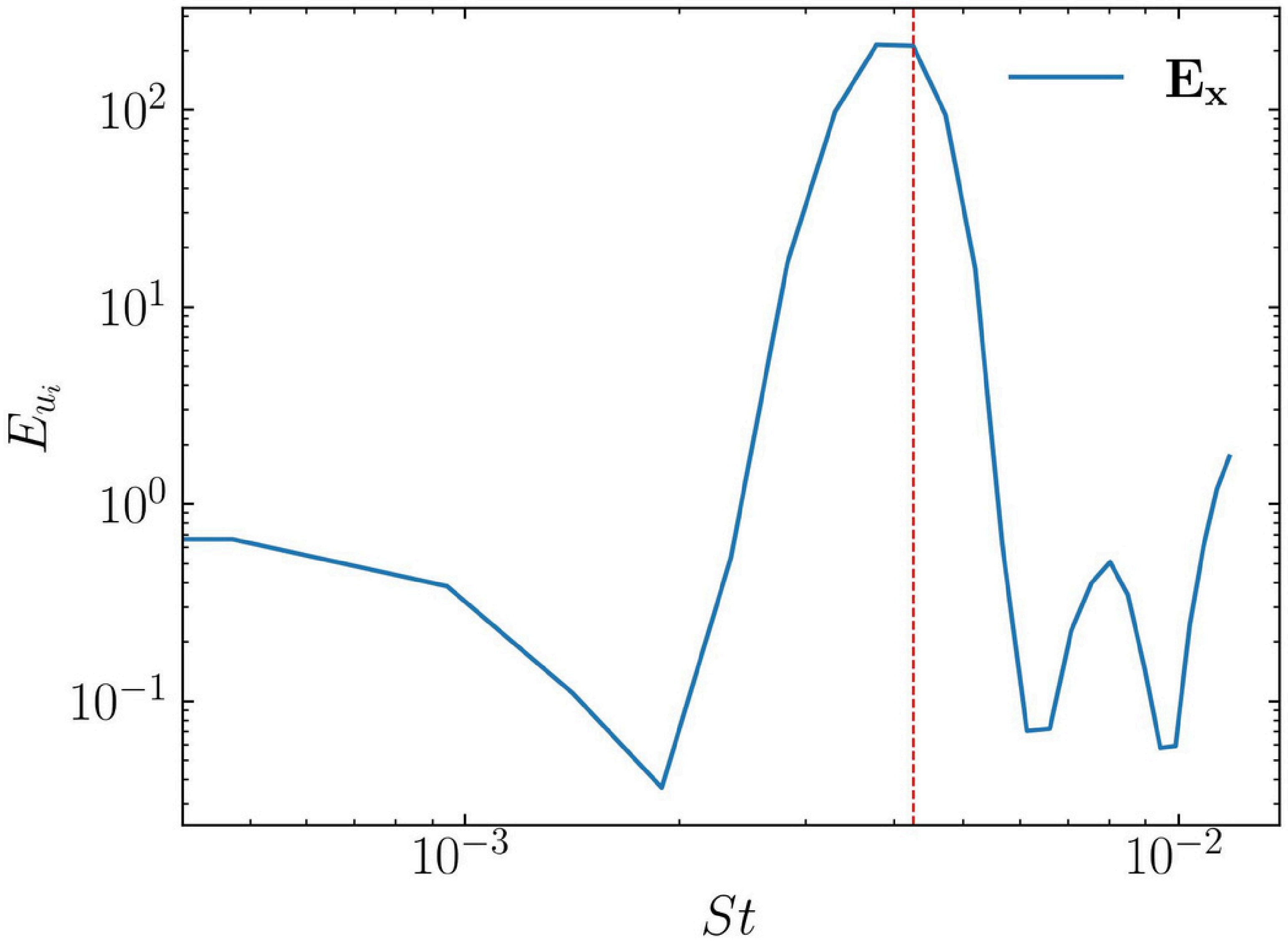
Strouhal Number from Welch = 0.004252
Dominant Frequency = 1.7647 Hz
Corresponding Period = 0.5667 s

{ St1 = f * d / Ub }

- f = frequency,
- d = characteristic length=0.04 m
- Ub = bulk velocity= 16.6m/s

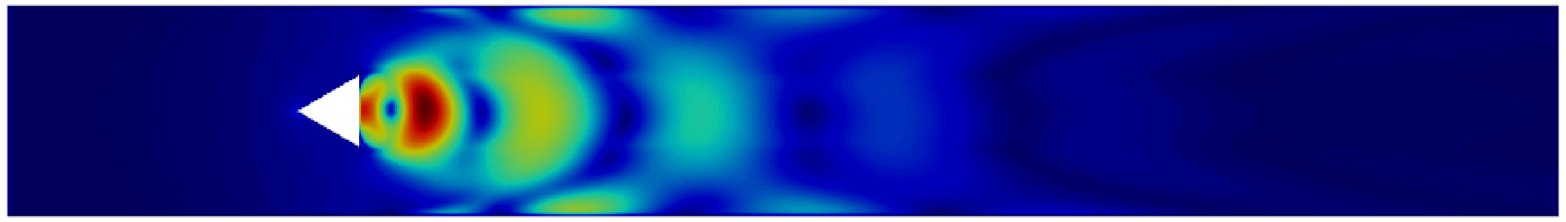


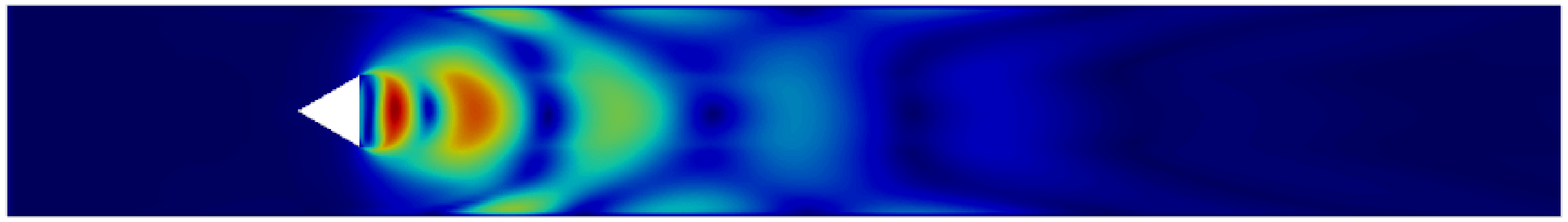
variation of u wrt time

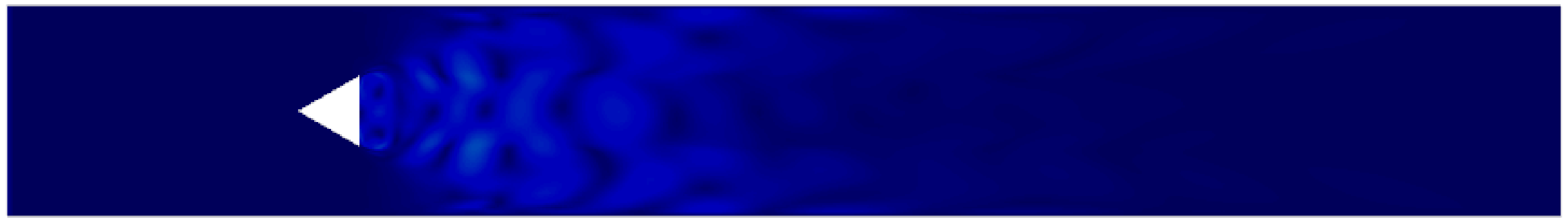


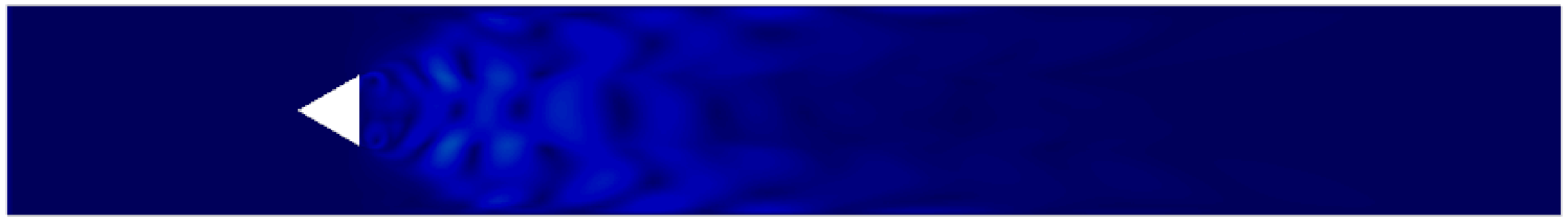
Power Spectral Density vs St. No.

First 10 modes


Mode 1


Mode 2


Mode 3

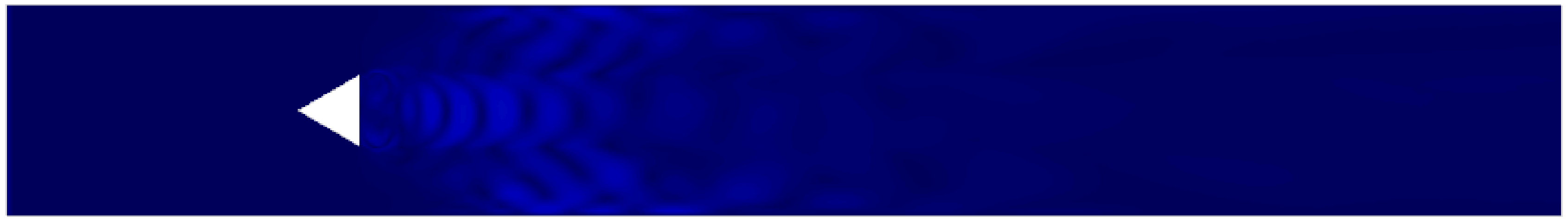

Mode 4

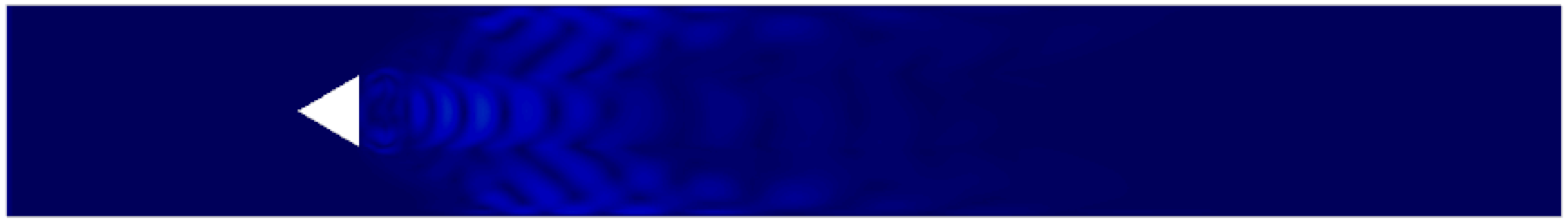

Mode 5


Mode 6

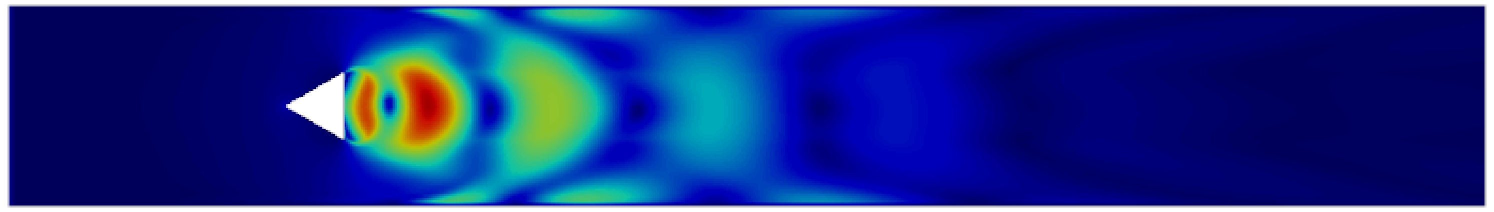

Mode 7

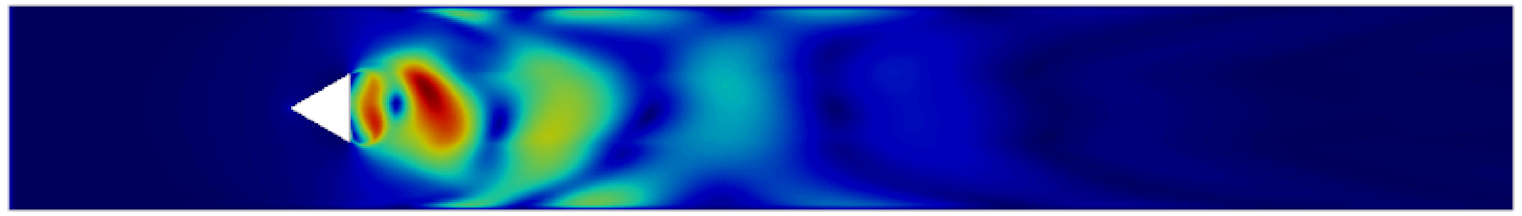

Mode 8

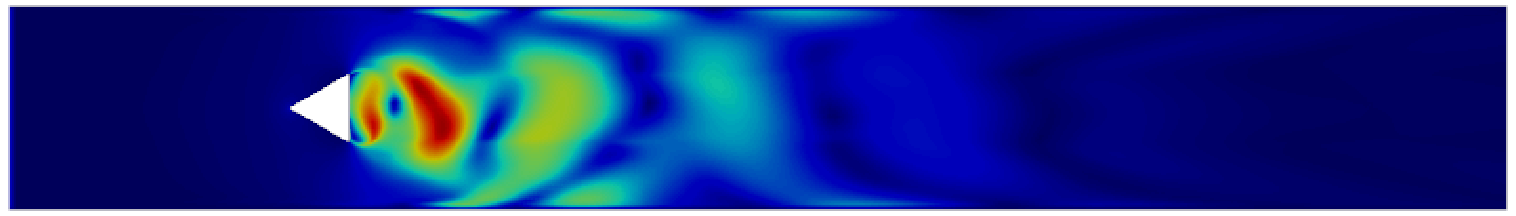

Mode 9

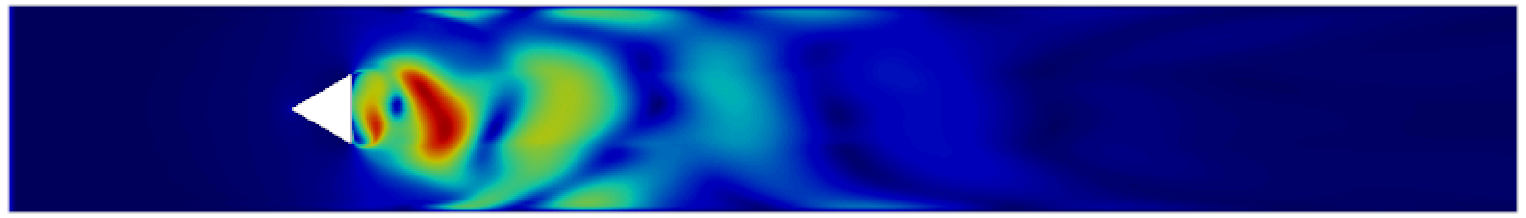

Mode 10

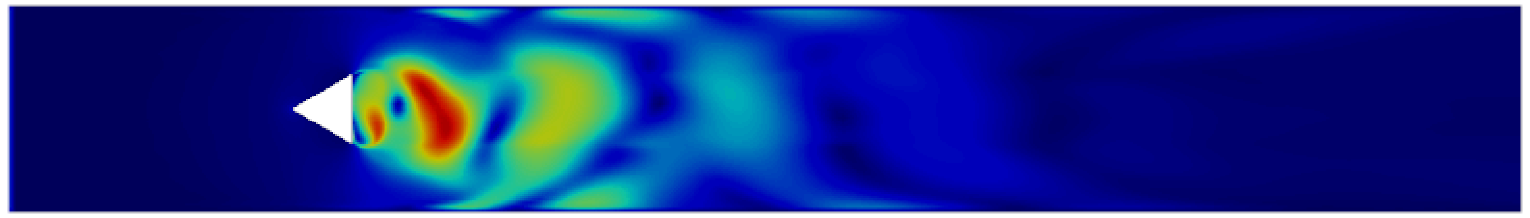
Reconstruction of flow :

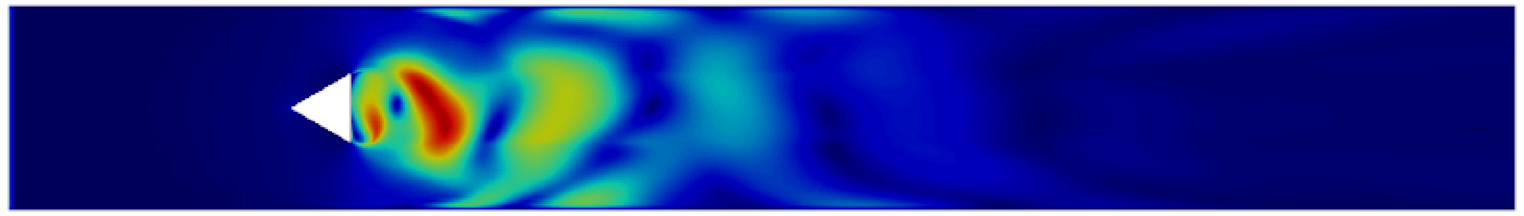

Mode 1+2

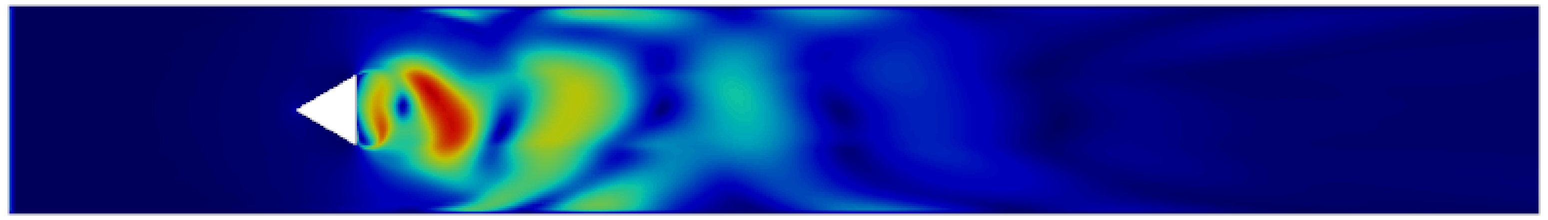

Mode 1+2+3

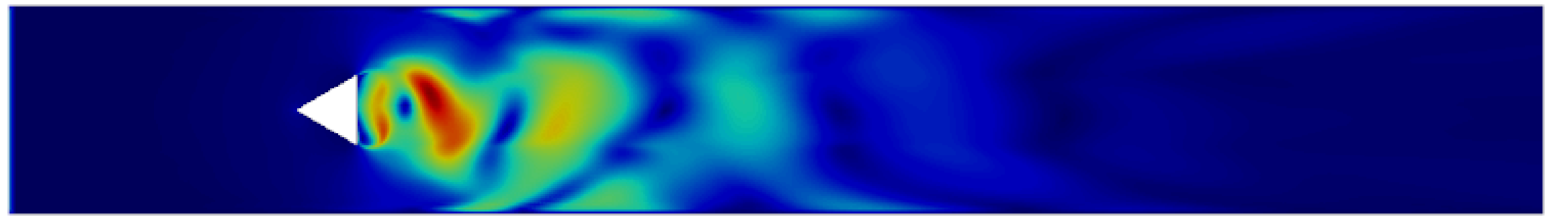

Mode 1+2+3+4

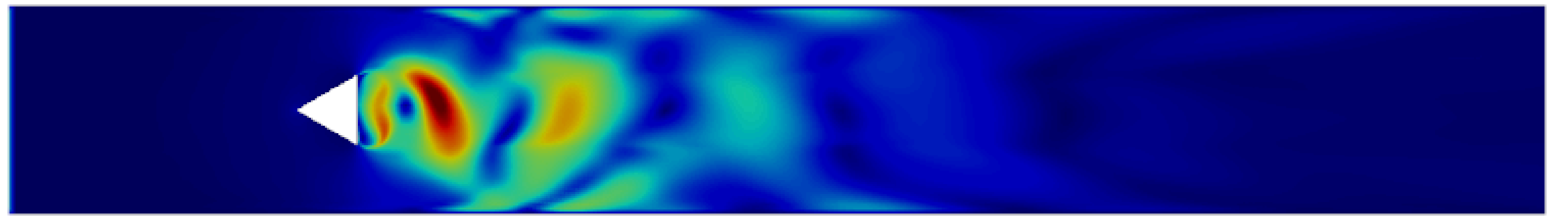

Mode 1+...+5


Mode 1+...+6


Mode 1+...+7

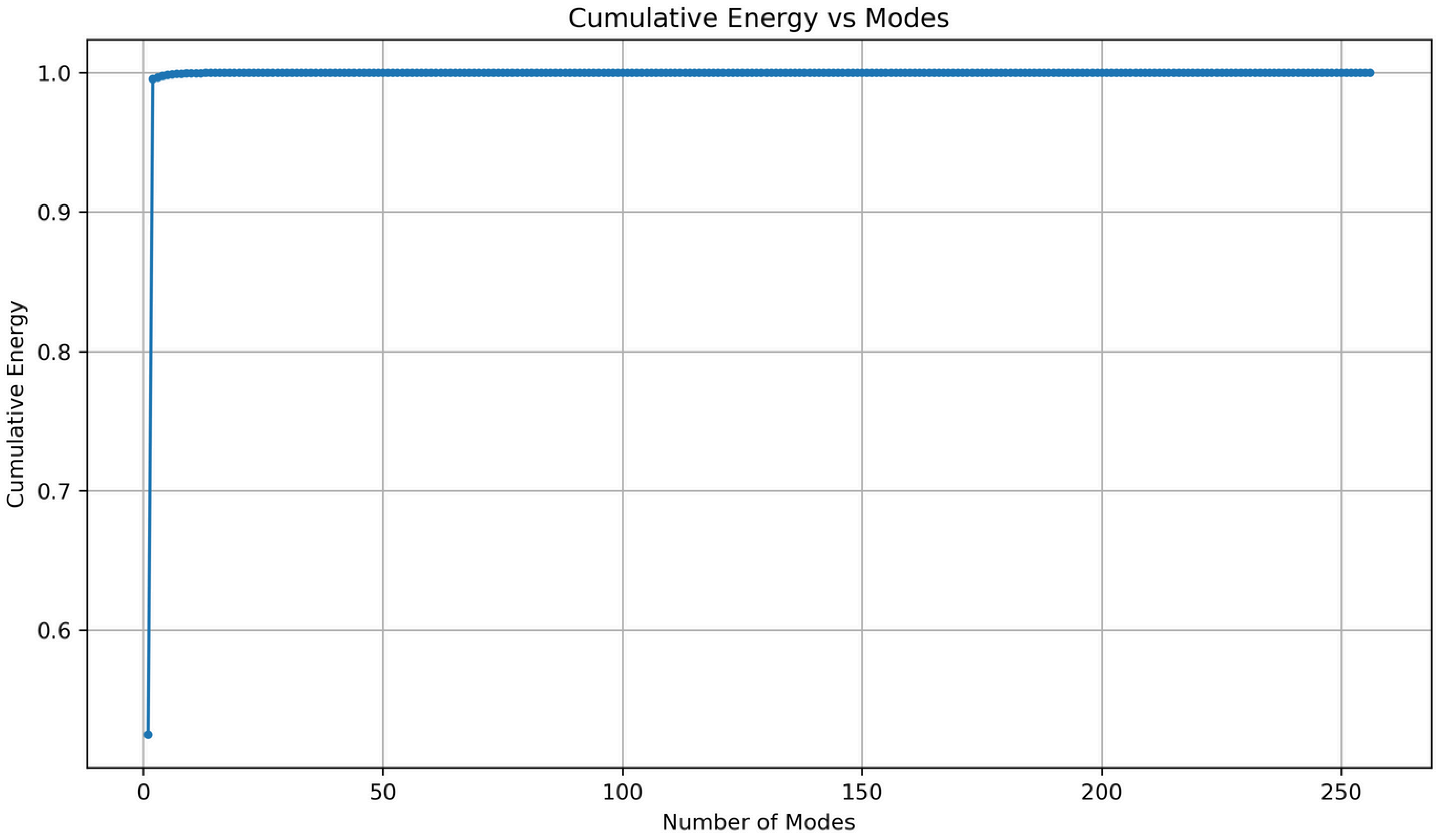

Mode 1+...+8

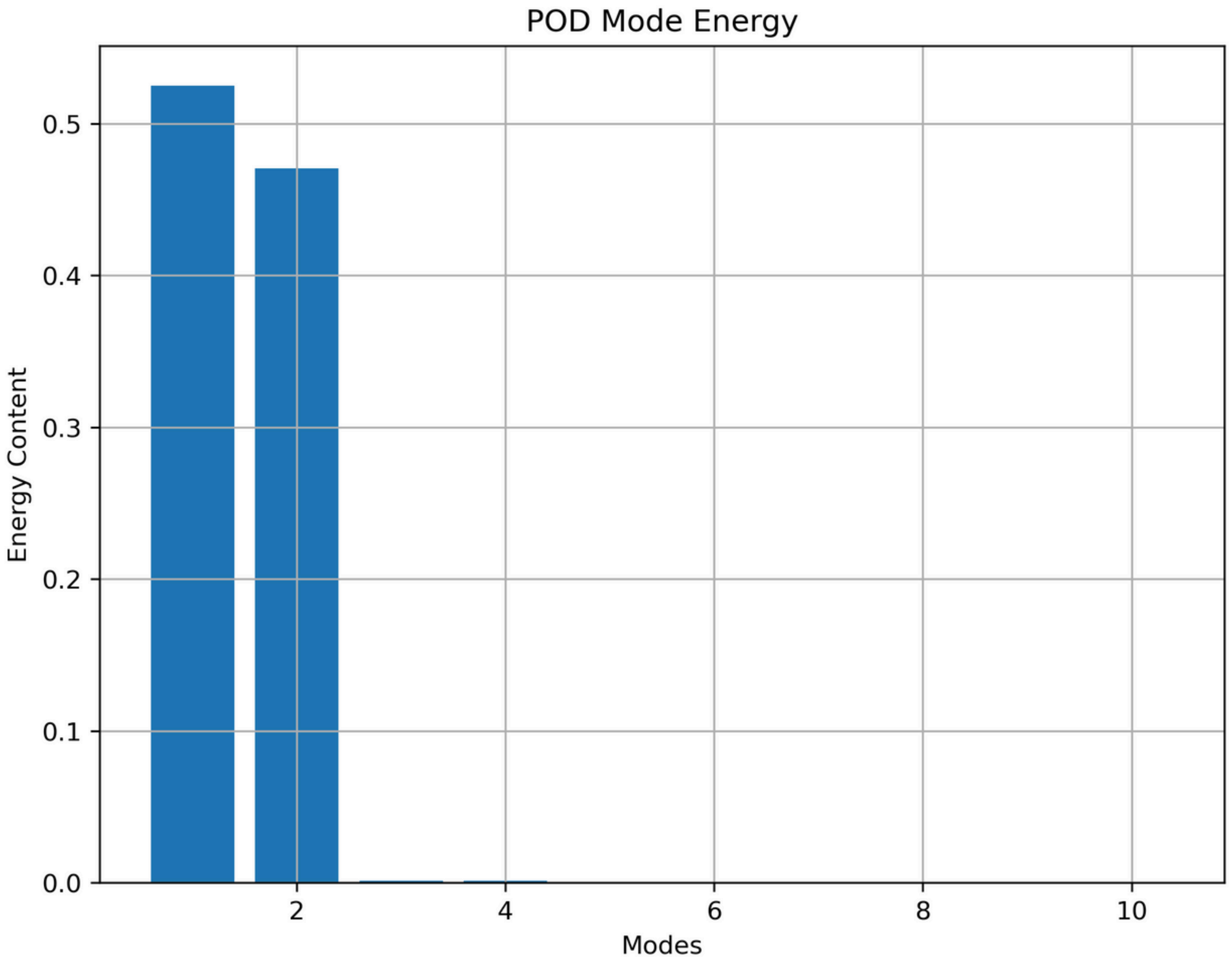

Mode 1+...+9


Mode 1+...+10

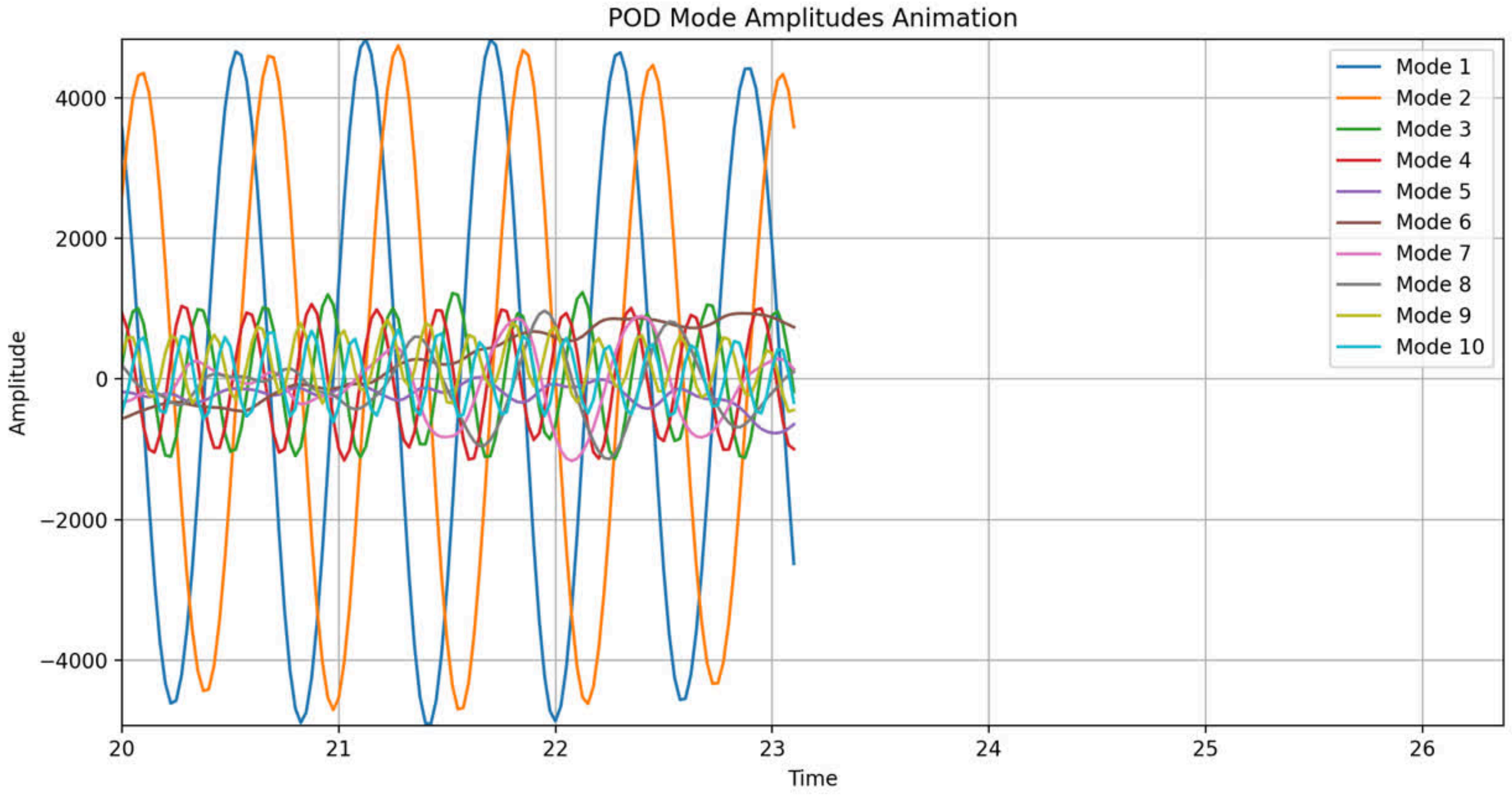
Energy Content of Modes

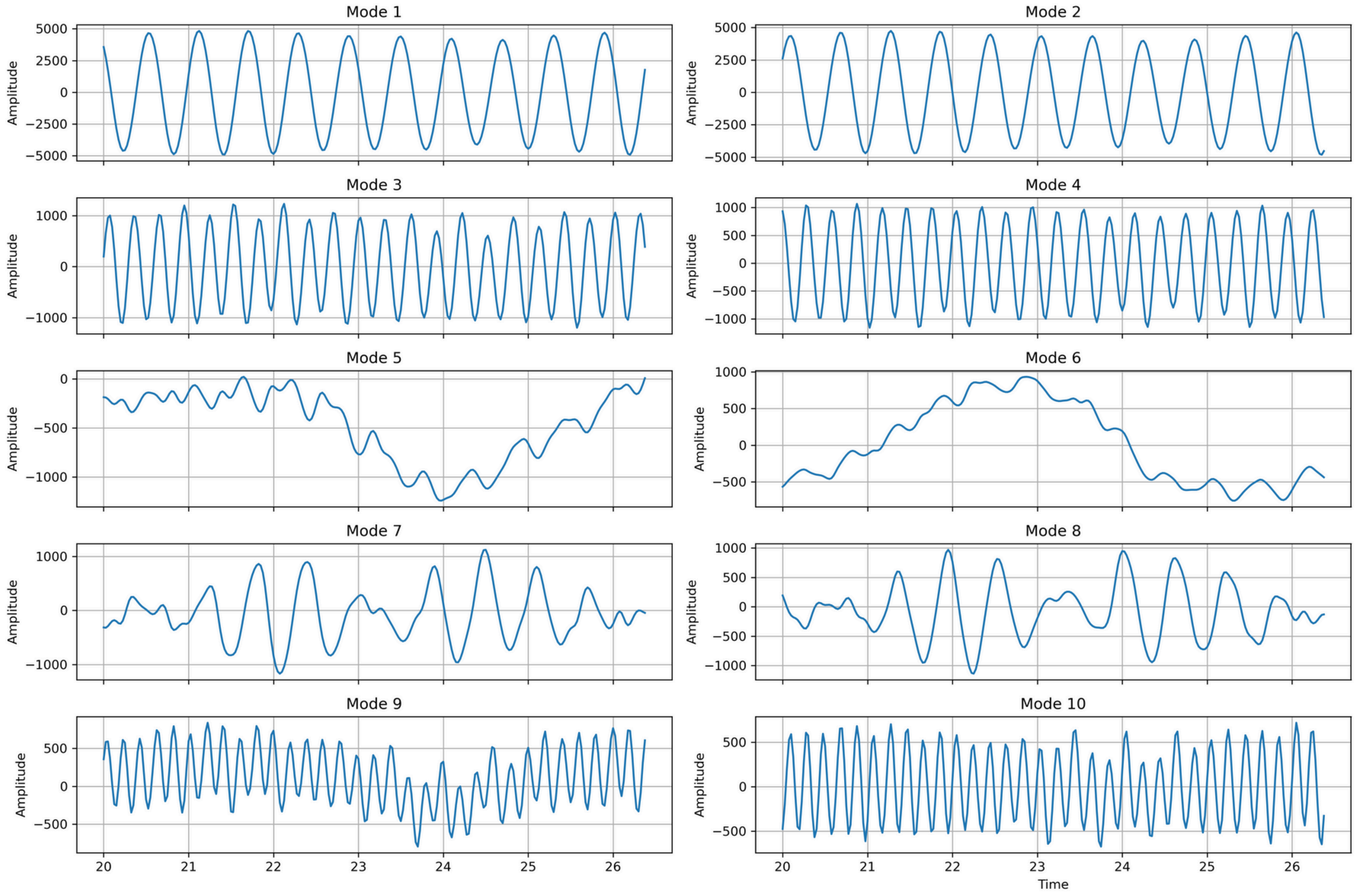
99% of Energy=first 2 modes
99.5% Energy=first 2 modes
99.7% Energy=First 7 modes





Amplitude variation wrt time :





Future Scope:

- POD results can be validated by Previous studies and experimental flow measurements (e.g., LDA/PIV data from the Volvo experiment).
- DMD can be performed to extract frequency-resolved coherent structures and compare with POD for dynamic insights.
- Grid Independence Study can be Conducted for grid convergence analysis to ensure results are mesh-independent despite current fine resolution.
- Adaptive Mesh Refinement (AMR) Sensitivity
  Analyze how refinement levels and vorticity thresholds impact flow features and POD mode accuracy.
- Extension to Reacting Flows
  Apply the same setup to simulate lean premixed combustion and study flame-turbulence interaction.
- Parametric Study of Inlet & Geometry
  Vary inlet velocity profile or bluff-body shape to observe their effects on flow instabilities and POD modes.